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Organic electroluminescent device including thin film encapsulation structure and method for producing same

Abstract

An organic EL device has a substrate, a drive circuit layer, a first inorganic protective layer, an organic planarizing layer, an organic EL element layer, and a TFE structure including a first inorganic barrier layer. When viewed from a normal line of the substrate, the organic planarizing layer is formed within a region where the first inorganic protective layer is formed, while an organic EL element is disposed within a region where the organic planarizing layer is formed. In a portion where the first inorganic protective layer and the first inorganic barrier layer are in direct contact with each other on a lead-out line of the drive circuit layer, the first inorganic barrier layer includes, in a cross-section parallel to a line width direction of the lead-out line, a lateral face having a taper angle of less than 90°.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) The present disclosure is a Continuation of the U.S. patent application Ser. No. 16/961,878, filed on Nov. 11, 2020, which is the National

TECHNICAL FIELD

(1) The present invention relates to an organic EL device (e.g., organic EL display device and organic EL illumination device) and a method for producing the same.

BACKGROUND ART

(2) Organic EL (Electroluminescent) display devices start being put into practical use. One feature of an organic EL display device is flexibility thereof. Such an organic EL display device includes, in each of pixels, at least one organic EL element (Organic Light Emitting Diode: OLED) and at least one TFT (Thin Film Transistor) controlling an electric current to be supplied to each of the at least one OLED. Hereinafter, an organic EL display device will be referred to as an “OLED display device”. Such an OLED display device including a switching element such as a TFT or the like for each of OLEDs is called an “active matrix OLED display device”. A substrate including the TFTs and the OLEDs will be referred to as an “element substrate”.

(3) An OLED (especially, an organic light emitting layer and a cathode electrode material) is easily influenced by moisture to be deteriorated and to cause display unevenness. One technology developed to provide an encapsulation structure that protects the OLED against moisture while not spoiling the flexibility of the OLED display device is a thin film encapsulation (TFE) technology. According to the thin film encapsulation technology, an inorganic barrier layer and an organic barrier layer are stacked alternately to allow such thin films to provide a sufficient level of water vapor barrier property. From the point of view of the moisture-resistance reliability of the OLED display device, such a thin film encapsulation structure is typically required to have a WVTR (Water Vapor Transmission Rate) lower than, or equal to, 1×10^{-4} g/m²/d ay.

(4) A thin film encapsulation structure used in OLED display devices commercially available currently includes an organic barrier layer (polymer barrier layer) having a thickness of about 5 μm to about 20 μm. Such a relatively thick organic barrier layer also has a role of flattening a surface of the element substrate. However, such a thick organic barrier layer involves a problem of limiting the bendability of the OLED display device.

(5) There is also a problem that the mass-productivity is low. The relatively thick organic barrier layer described above is formed by use of a printing technology such as an inkjet method, a microjet method or the like. By contrast, an inorganic barrier layer is formed by a thin film formation technology in a vacuum atmosphere (e.g., lower than, or equal to, 1 Pa). The formation of the organic barrier layer by use of a printing method is performed in the air or a nitrogen atmosphere, whereas the formation of the inorganic barrier layer is performed in vacuum. Therefore, the element substrate is put into, and out of, a vacuum chamber during the formation of the thin film encapsulation structure, which decreases the mass-productivity.

(6) Under such a situation, as disclosed in, for example, Patent Document No. 1, a film formation device capable of producing an inorganic barrier layer and an organic barrier layer continuously has been developed.

(7) Patent Document No. 2 discloses a thin film encapsulation structure including a first inorganic material layer, a first resin member and a second inorganic material layer provided on the element substrate in this order, with the first inorganic material layer being closest to the element substrate. In this thin film encapsulation structure, the first resin member is present locally, more specifically, in the vicinity of a protruding portion of the first inorganic material layer (first inorganic material layer covering the protruding portion). According to Patent Document No. 2, since the first resin member is present locally, more specifically, in the vicinity of the protruding portion, which may not be sufficiently covered with the first inorganic material layer, entrance of moisture or oxygen via the non-covered portion is suppressed. In addition, the first resin member acts as an underlying layer for the second inorganic material layer. Therefore, the second inorganic material layer is

properly formed and properly covers a side surface of the first inorganic material layer with an expected thickness. The first resin member is formed as follows. An organic material heated and vaporized to be mist-like is supplied onto the element substrate maintained at a temperature lower than, or equal to, room temperature. The organic material is condensed and put into liquid drops on the substrate. The organic material in the liquid drops moves on the substrate by a capillary action or a surface tension to be present locally, more specifically, at a border between a side surface of the protruding portion of the first inorganic material layer and a surface of the substrate. Then, the organic material is cured to form the first resin member at the border. Patent Document No. 3 also discloses an OLED display device including a similar thin film encapsulation structure. Patent Document No. 4 discloses a film formation device usable to produce an OLED display device.

CITATION LIST

Patent Literature

(8) Patent Document No. 1: Japanese Laid-Open Patent Publication No. 2013-186971 Patent Document No. 2: WO2014/196137 Patent Document No. 3: Japanese Laid-Open Patent Publication No. 2016-39120 Patent Document No. 4: Japanese Laid-Open Patent Publication No. 2013-64187

SUMMARY OF INVENTION

Technical Problem

(9) The thin film encapsulation structure that is described in each of Patent Documents Nos. 2 and 3 does not include a thick organic barrier layer. Therefore, the thin film encapsulation structure is considered to improve the bendability of the OLED display device. In addition, since the inorganic barrier layer and the organic barrier layer may be formed continuously, the mass-productivity is also improved.

(10) However, according to the studies made by the present inventors, an organic barrier layer formed by the method described in Patent Document No. 2 or 3 has a problem of not providing a sufficiently high level of moisture-resistance reliability.

(11) In the case where an organic barrier layer is formed by use of a printing method such as an inkjet method or the like, it is possible to form the organic barrier layer only in an active region on the element substrate (the active region may also be referred to as an “element formation region” or a “display region”) but not in a region other than the active region. In this case, along a periphery of the active region (outer to the active region), there is a region where the first inorganic material layer and the second inorganic material layer are in direct contact with each other, and the organic barrier layer is fully enclosed by the first inorganic material layer and the second inorganic material layer and is isolated from the outside of the first inorganic material layer and the second inorganic material layer.

(12) By contrast, according to the method for forming the organic barrier layer described in Patent Document No. 2 or 3, a resin (organic material) is supplied to the entire surface of the element substrate, and the surface tension of the resin, which is in a liquid state, is used to locate the resin locally, more specifically, at the border between the surface of the element substrate and the side surface of the protruding portion on the surface of the element substrate. Therefore, the organic barrier layer may also be formed in a region other than the active region (the region other than the active region may also be referred to as a “peripheral region”), namely, a terminal region where a plurality of terminals are located and a lead wire region where lead wires extending from the active region to the terminal region are formed. Specifically, the resin is present locally, more specifically, at, for example, the border between side surfaces of the lead wires or side surfaces of the terminals and the surface of the substrate. In this case, an end of a portion, of the organic barrier layer, that is formed along the lead wires is not enclosed by the combination of the first inorganic barrier layer and the second inorganic barrier layer, but is exposed to the air (ambient atmosphere).

(13) The organic barrier layer is lower in the water vapor barrier property than the inorganic barrier layer. Therefore, the organic barrier layer formed along the lead wires acts as a route that leads

water vapor in the air to the active region.

(14) In the above, some problems of a thin film encapsulation structure preferably usable for a flexible organic EL display device are described. The thin film encapsulation structure is not limited to being used for an organic EL display device, but is also usable for other types of organic EL devices such as an organic EL illumination device and the like.

(15) The present invention, made to solve the above-described problems, has an object of providing an organic EL device including a thin film encapsulation structure that includes a relatively thin organic barrier layer, by which the mass-productivity and the moisture-resistance reliability are improved, and also providing a method for producing the same.

Solution to Problem

(16) An organic EL device according to an embodiment of the present invention includes a substrate; a driving circuit layer including a plurality of TFTs formed on the substrate, a plurality of gate bus lines and a plurality of source bus lines each connected with either one of the plurality of TFTs, a plurality of terminals, and a plurality of lead wires each connecting either one of the plurality of terminals and either one of the plurality of gate bus lines or either one of the plurality of source bus lines to each other; a first inorganic protective layer formed on the driving circuit layer and exposing at least the plurality of terminals; an organic flattening layer formed on the first inorganic protective layer and having a surface that has an arithmetic average roughness of greater than 50 nm; a second inorganic protective layer formed on the organic flattening layer and having a surface that has an arithmetic average roughness of nm or less; an organic EL element layer including a plurality of organic EL elements formed on the second inorganic protective layer and each connected with either one of the plurality of TFTs; and a thin film encapsulation structure formed so as to cover the organic EL element layer and including a first inorganic barrier layer, an organic barrier layer in contact with a top surface of the first inorganic barrier layer and a second inorganic barrier layer in contact with a top surface of the organic barrier layer, the organic barrier layer being formed in a region enclosed by an inorganic barrier layer joint portion where the first inorganic barrier layer and the second inorganic barrier layer are in direct contact with each other. As viewed in a direction of normal to the substrate, the organic flattening layer is formed in a region where the first inorganic protective layer is formed, the plurality of organic EL elements are located in a region where the organic flattening layer is formed, an outer perimeter of the thin film encapsulation structure crosses the plurality of lead wires and is present between an outer perimeter of the organic flattening layer and an outer perimeter of the first inorganic protective layer. In a portion where the first inorganic protective layer and the first inorganic barrier layer are in direct contact with each other on the plurality of lead wires, a side surface of a cross-sectional shape, of the first inorganic barrier layer, that is parallel to a line width direction of the plurality of lead wires has a tapering angle of less than 90 degrees.

(17) In an embodiment, the tapering angle of the side surface of the first inorganic barrier layer is less than 70 degrees.

(18) In an embodiment, the organic flattening layer is formed of a non-photosensitive resin.

(19) In an embodiment, the organic flattening layer is formed of polyimide.

(20) A method for producing an organic EL device according to an embodiment of the present invention is a method for producing the organic EL device of any one of the above. The method includes step A of forming the driving circuit layer on the substrate; step B of forming a first inorganic protective film on the driving circuit layer; step C of forming an organic flattening film on the first inorganic protective film; step D of forming a second inorganic protective film on the organic flattening film; step E of treating a surface of the second inorganic protective film with chemical mechanical polishing; step F of patterning the first inorganic protective film, the organic flattening film and the second inorganic protective film to form the first inorganic protective layer, the organic flattening layer and the second inorganic protective layer; step G of baking the organic flattening layer to a temperature of 100° C. or higher; and step H of, after the step G, forming an

organic layer on the second inorganic protective layer, the organic layer being contained in the organic EL elements.

(21) In an embodiment, the step F includes the step of forming a contact hole running through the first inorganic protective film, the organic flattening film and the second inorganic protective film in one step.

(22) In an embodiment, the method further includes: after the step F but before the step G, step F1 of forming a positive photoresist film covering the organic flattening layer, and step F2 of exposing the entirety of the photoresist film to light and then developing the photoresist film to remove the photoresist film.

(23) In an embodiment, the method further includes the step of, between the step F1 and the step F2, storing or transporting the substrate having the photoresist film formed thereon.

(24) In an embodiment, the method further includes step I of, after the step H, forming the first inorganic barrier layer selectively on an active region where the plurality of organic EL elements are formed; step J of, after the step I, locating the substrate in a chamber and supplying vapor of a photocurable resin or a mist-like photocurable resin into the chamber; step K of condensing the photocurable resin on the first inorganic barrier layer such that the photocurable resin is not present on a portion, of the first inorganic barrier layer, that has the tapering angle of less than 90 degrees; and step L of, after the step K, irradiating the condensed photocurable resin with light to form the organic barrier layer of a photocured resin.

(25) In an embodiment, the method further includes step I of, after the step H, forming the first inorganic barrier layer selectively on an active region where the plurality of organic EL elements are formed; step J of, after the step I, locating the substrate in a chamber and supplying vapor of a photocurable resin or a mist-like photocurable resin into the chamber; step K of condensing the photocurable resin on the first inorganic barrier layer to form a liquid film; step L of irradiating the liquid film of the photocurable resin with light to form a photocured resin layer; and step M of partially asking the photocured resin layer to form the organic barrier layer.

Advantageous Effects of Invention

(26) An embodiment of the present invention provides an organic EL device including a thin film encapsulation structure that includes a relatively thin organic barrier layer, by which the mass-productivity and the moisture-resistance reliability are improved, and also provides a method for producing the same.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1A is a schematic partial cross-sectional view of an active region of an OLED display device **100** according to an embodiment of the present invention, and FIG. 1B is a partial cross-sectional view of a TFE structure **10** formed on an OLED **3**.

(2) FIG. 2 is a schematic plan view of an OLED display device **100A** according to an embodiment of the present invention.

(3) FIG. 3A and FIG. 3B are each a schematic cross-sectional view of the OLED display device **100A**; FIG. 3A is a cross-sectional view taken along line 3A-3A' in FIG. 2, and FIG. 3B is a cross-sectional view taken along line 3B-3B' in FIG. 2; and FIG. 3C is a cross-sectional view showing tapering angles $\square$ of side surfaces of layers.

(4) FIG. 4A through FIG. 4D are each a schematic cross-sectional view of the OLED display device **100A**; FIG. 4A is a cross-sectional view taken along line 4A-4A' in FIG. 2, FIG. 4B is a cross-sectional view taken along line 4B-4B' in FIG. 2, FIG. 4C is a cross-sectional view taken along line 4C-4C' in FIG. 2, and FIG. 4D is a cross-sectional view taken along line 4D-4D' in FIG. 2.

- (5) FIG. 5A and FIG. 5B are respectively schematic cross-sectional views of OLED display devices **100B1** and **100B2** according to comparative examples, each corresponding to FIG. 4B.
- (6) FIG. 6 is a schematic plan view of an OLED display device **100C** according to a comparative example.
- (7) FIG. 7A and FIG. 7B are each a schematic cross-sectional view of the OLED display device **100C**; FIG. 7A is a cross-sectional view taken along line 7A-7A' in FIG. 6, and FIG. 7B is a cross-sectional view taken along line 7B-7B' in FIG. 6.
- (8) FIG. 8A through FIG. 8C are each a schematic cross-sectional view of the OLED display device **100C**; FIG. 8A is a cross-sectional view taken along line 8A-8A' in FIG. 6, FIG. 8B is a cross-sectional view taken along line 8B-8B' in FIG. 6, and FIG. 8C is a cross-sectional view taken along line 8C-8C' in FIG. 6.
- (9) FIG. 9A and FIG. 9B are each a schematic cross-sectional view showing an example of TFT that may be included in an OLED display device according to an embodiment.
- (10) FIG. 10A through FIG. 10C are schematic cross-sectional views of another OLED display device according to an embodiment, respectively corresponding to FIG. 4B through FIG. 4D.
- (11) FIG. 11A through FIG. 11C are schematic cross-sectional views of still another OLED display device according to an embodiment.
- (12) FIG. 12A and FIG. 12B schematically show a structure of a film formation device **200**; FIG. 12A shows a state of the film formation device **200** in a step of condensing a photocurable resin on a first inorganic barrier layer, and FIG. 12B shows a state of the film formation device **200** in a step of curing the photocurable resin.

DESCRIPTION OF EMBODIMENTS

- (13) Hereinafter, an OLED display device and a method for producing the same according to an embodiment of the present invention will be described with reference to the drawings. In the following, an OLED display device including a flexible substrate will be described as an example. Nonetheless, embodiments of the present invention are not limited to being directed to an organic EL display device, but may be directed to another organic EL device such as an organic EL illumination device or the like. Embodiments of the present invention are not limited to the embodiments described below as examples.
- (14) First, with reference to FIG. 1A and FIG. 1B, a basic structure of an OLED display device **100** according to an embodiment of the present invention will be described. FIG. 1A is a schematic partial cross-sectional view of an active region of the OLED display device **100** according to an embodiment of the present invention. FIG. 1B is a schematic partial cross-sectional view of a TFE structure **10** formed on an OLED **3**.
- (15) The OLED display device **100** includes a plurality of pixels, and each of the pixels includes at least one organic EL element (OLED). Herein, a structure corresponding to one OLED will be described for the sake of simplicity.
- (16) As shown in FIG. 1A, the OLED display device **100** includes a flexible substrate (hereinafter, may be referred to simply as a “substrate”) **1**, a circuit (may be referred to as a “driving circuit” or a “backplane”) **2** formed on the substrate **1** and including a TFT, an inorganic protective layer (first inorganic protective layer) **2Pa** formed on the circuit **2**, an organic flattening layer **2Pb** formed on the inorganic protective layer **2Pa**, the OLED **3** formed on the organic flattening layer **2Pb**, and the TFE structure **10** formed on the OLED **3**. Hereinafter, an assembly of the substrate **1**, the circuit **2** formed on the substrate **1**, the inorganic protective layer **2Pa**, the organic flattening layer **2Pb** and the OLED **3** may be referred to as an “element substrate **20**”. Namely, the assembly of the substrate **1** and the components that are formed on the substrate **1** and are closer to the substrate **1** than the TFE structure **10** may be referred to as the “element substrate”.
- (17) The OLED **3** is, for example, of a top emission type. An uppermost portion of the OLED **3** is, for example, an upper electrode or a cap layer (refractive index adjusting layer). A layer including a plurality of the OLEDs **3** may be referred to as an “OLED layer **3**”. An optional polarizing plate **4**

is located on the TFE structure **10**. The circuit **2** and the OLED layer **3** may share a part of the components. For example, a layer having a touch panel function may be located between the TFE structure **10** and the polarizing plate **4**. Namely, the OLED display device **100** may be altered to a display device including an on-cell type touch panel.

(18) The substrate **1** is, for example, a polyimide film having a thickness of 15 μm . The circuit **2** including the TFT has a thickness of, for example, 4 μm . The inorganic protective layer **2Pa** has a structure of, for example, SiN.sub.x layer (500 nm)/SiO.sub.2 layer (100 nm) (top layer/bottom layer). Alternatively, the inorganic protective layer **2Pa** may have a three-layer structure of, for example, SiO.sub.2 layer/SiN.sub.x layer/SiO.sub.2 layer. These layers may have thicknesses of, for example, 200 nm/300 nm/100 nm. Still alternatively, the inorganic protective layer **2Pa** may be a single layer of SiN.sub.x (200 nm). The organic flattening layer **2Pb** is, for example, an acrylic resin layer, an epoxy resin layer or a polyimide layer having a thickness of 4 μm . The organic flattening layer **2Pb** is preferably formed of a non-photosensitive resin, but may be formed of a photosensitive resin. The OLED **3** has a thickness of, for example, 1 μm . The TFE structure **10** has a thickness of, for example, 2.5 μm or less.

(19) FIG. **13** is a partial cross-sectional view of the TFE structure **10** formed on the OLED **3**. A first inorganic barrier layer (e.g., SiN.sub.x layer) **12** is formed immediately on the OLED **3**, an organic barrier layer (e.g., acrylic resin layer) **14** is formed on the first inorganic barrier layer **12**, and a second inorganic barrier layer (e.g., SiN.sub.x layer) **16** is formed on the organic barrier layer **14**.

(20) For example, the first inorganic barrier layer **12** is an SiN.sub.x layer having a thickness of, for example, 1.5 μm , the second inorganic barrier layer **16** is an SiN.sub.x layer having a thickness of, for example, 800 nm, and the organic barrier layer **14** is an acrylic resin layer having a thickness of, for example, less than 100 nm. Each of the first inorganic barrier layer **12** and the second inorganic barrier layer **16** independently has a thickness of 200 nm or greater and 1500 nm or less. The organic barrier layer **14** has a thickness of 50 nm or greater and less than 200 nm. The TFE structure has a thickness of, preferably, 400 nm or greater and less than 3 μm , and more preferably, 400 nm or greater and 2.5 μm or less.

(21) The TFE structure **10** is formed to protect the active region (see the active region R1 in FIG. **2**) of the OLED display device **100**. As described above, the TFE structure **10** includes, on the active region R1, the first inorganic barrier layer **12**, the organic barrier layer **14** and the second inorganic barrier layer **16** sequentially provided with the first inorganic barrier layer **12** being closest to the OLED **3**. The organic barrier layer **14** is not present as a film covering the entirety of the active region R1, but has an opening. A portion, of the organic barrier layer **14**, that excludes the opening, namely, a portion where an organic film is actually present, will be referred to as a “solid portion”. The organic barrier layer **14** may be formed by the method described in, for example, Patent Document No. 1 or 2 or by use of a film formation device **200** described below.

(22) The “opening” (may also be referred to as a “non-solid portion”) does not need to be enclosed by the solid portion, but includes a cutout portion or the like. In the opening, the first inorganic barrier layer **12** and the second inorganic barrier layer **16** are in direct contact with each other. Hereinafter, a portion where the first inorganic barrier layer **12** and the second inorganic barrier layer **16** are in direct contact with each other will be referred to as an “inorganic barrier layer joint portion”.

(23) Now, with reference to FIG. **2** and FIG. **3**, a structure of, and a method for producing, an OLED display device **100A** according to an embodiment of the present invention will be described.

(24) FIG. **2** shows a schematic plan view of the OLED display device **100A** according to an embodiment of the present invention. With reference to FIG. **3A** through FIG. **3C** and FIG. **4A** through FIG. **4D**, a cross-sectional structure of the OLED display device **100A** will be described. FIG. **3A** and FIG. **3B** are each a schematic cross-sectional view of the OLED display device **100A**. FIG. **3A** is a cross-sectional view taken along line 3A-3A' in FIG. **2**, and FIG. **3B** is a cross-sectional view taken along line 3B-3B' in FIG. **2**. FIG. **3C** is a cross-sectional view showing

tapering angles □ of side surfaces of layers. FIG. 4A through FIG. 4D are each a schematic cross-sectional view of the OLED display device **100A**. FIG. 4A is a cross-sectional view taken along line **4A-4A'** in FIG. 2. FIG. 4B is a cross-sectional view taken along line **4B-4B'** in FIG. 2. FIG. 4C is a cross-sectional view taken along line **4C-4C'** in FIG. 2. FIG. 4D is a cross-sectional view taken along line **4D-4D'** in FIG. 2.

(25) First, FIG. 2 will be referred to. The circuit **2** formed on the substrate **1** includes a plurality of TFTs (not shown), and a plurality of gate bus lines (not shown) and a plurality of source bus lines (not shown) each connected to either one of the plurality of TFTs (not shown). The circuit **2** may be a known circuit that drives the plurality of OLEDs **3**. The plurality of OLEDs **3** are each connected with either one of the plurality of TFTs included in the circuit **2**. The OLEDs **3** may be known OLEDs.

(26) The circuit **2** further includes a plurality of terminals **34** located in a peripheral region **R2** outer to the active region (region enclosed by the dashed line in FIG. 2) **R1**, where the plurality of OLEDs **3** are located, and also includes a plurality of lead wires **32** each connecting either one of the plurality of terminals **34** and either one of the plurality of gate bus lines or either one of the plurality of source bus lines to each other. The entirety of the circuit **2** including the plurality of TFTs, the plurality of gate bus lines, the plurality of source bus lines, the plurality of lead wires **32** and the plurality of terminals **34** may be referred to as a “driving circuit layer **2**”. A portion, of the driving circuit layer **2**, that is formed in the active region **R1** will be referred to as a “driving circuit layer **2A**”.

(27) In FIG. 2 and the like, only the lead wires **32** and/or the terminals **34** may be shown as components of the driving circuit layer **2**. Nonetheless, the driving circuit layer **2** includes a conductive layer including the lead wires **32** and the terminals **34** and further includes at least one conductive layer, at least one insulating layer, and at least one semiconductor layer. The structure of the conductive layers, the insulating layer and the semiconductor layer included in the driving circuit layer **2** may be changed in accordance with the structure of the TFT shown in, for example, FIG. 9A or FIG. 9B as an example. An insulating film (base coat) may be formed on the substrate **1** as an underlying layer for the driving circuit layer **2**.

(28) As viewed in a direction of normal to the substrate **1**, the organic flattening layer **2Pb** is formed in a region where the inorganic protective layer **2Pa** is formed. The active region **R1** (**2A**, **3**) is formed in a region where the organic flattening layer **2Pb** is formed. An outer perimeter of the TFE structure **10** crosses the plurality of lead wires **32**, and is present between an outer perimeter of the organic flattening layer **2Pb** and an outer perimeter of the inorganic protective layer **2Pa**. Therefore, the organic flattening layer **2Pb** is enclosed, together with the OLED layer **3**, by a joint portion where the inorganic protective layer **2Pa** and the first inorganic barrier layer **12** are in direct contact with each other (see FIG. 3B and FIG. 4B). The inorganic protective layer **2Pa** is formed by, for example, plasma CVD using a mask (e.g., metal mask) so as to expose at least the plurality of terminals **34**. Alternatively, the inorganic protective layer **2Pa** may be formed as follows, for example. An inorganic protective film is formed by plasma CVD so as to cover the terminals **34** and then is patterned by a photolithography process (including a dry etching step), so that an opening exposing the terminals **34** is formed. As described below, in the case where a second inorganic protective layer **2Pa2** is formed on the organic flattening layer **2Pb** (see FIG. 11B), it is preferred that the patterning including the formation of a contact hole is performed in one step for an inorganic protective film **2Pa** (first inorganic protective film **2Pa1**), an organic flattening film **2Pb** and the second inorganic protective film **2Pa2**.

(29) The inorganic protective layer **2Pa** protects the driving circuit layer **2**. The organic flattening layer **2Pb** flattens a surface of an underlying layer on which the OLED layer **3** is to be formed. Like the organic barrier layer **14**, the organic flattening layer **2Pb** has a water vapor barrier property lower than that of the inorganic protective layer **2Pa** or the inorganic barrier layer **12** or **16**.

Therefore, as shown in FIG. 6 through FIG. 8 regarding an OLED display device **100C** according

to a comparative example, in the case where an organic flattening layer 2Pbc is partially exposed to the air (ambient atmosphere), the organic flattening layer 2Pbc absorbs moisture from the exposed portion. As a result, the organic flattening layer 2Pbc acts as a route that guides the moisture in the air into the active region R1. As described above, in the OLED display device 100 according to an embodiment, the organic flattening layer 2Pb is enclosed by the joint portion where the inorganic protective layer 2Pa and the first inorganic barrier layer 12 are in direct contact with each other. Therefore, the organic flattening layer 2Pb is prevented from guiding the moisture into the active region R1.

(30) The organic flattening layer 2Pb is formed of an organic resin. As the organic resin, an acrylic resin, an epoxy resin or a polyimide resin is preferably usable. The organic flattening layer 2Pb is obtained by patterning an organic resin film formed by any of various application methods or printing methods. The organic resin film is formed of, for example, a non-photosensitive resin. It should be noted that a photosensitive resin may be used. It is preferred that a positive photosensitive resin is used as the photosensitive resin. Use of a photosensitive resin allows the organic resin film to be formed only in a predetermined region by a photolithography process. For example, a photosensitive film is applied onto substantially the entirety of the element substrate and then is exposed to light and developed, so that the organic resin film may be formed in a predetermined region excluding a peripheral region of the element substrate (see, for example, FIG. 11C).

(31) Regarding the patterning of the organic resin film, in the case where the second inorganic protective layer 2Pa2 is formed on the organic flattening layer 2Pb (see FIG. 11B), it is preferred that the contact hole is formed in one step in the inorganic protective film 2Pa (first inorganic protective film 2Pa1), the organic flattening film 2Pb and the second inorganic protective film 2Pa2.

(32) An organic resin easily absorbs moisture. Meanwhile, as described above, the OLED (especially, an organic light emitting layer and a cathode electrode material) is easily influenced by moisture to be deteriorated. Therefore, it is preferred that before the OLED layer 3 is formed on the organic flattening layer 2Pb, baking is performed in order to remove the moisture contained in the organic flattening layer 2Pb. The baking temperature is preferably, for example, 100° C. or higher (e.g., for 1 hours or longer), and more preferably 250° C. or higher (e.g., for 15 minutes or longer). The atmosphere may be an atmospheric pressure. Baking in a low pressure atmosphere may shorten the baking time. It is preferred to use a resin material having a high thermal resistance, for example, polyimide, in order to prevent deterioration by heat in the baking step.

(33) There are cases where after the organic flattening layer 2Pb is formed but before the OLED layer 3 is formed, the element substrate is stored or transported. Namely, after the element substrate including the driving circuit layer 2, the inorganic protective layer 2Pa and the organic flattening layer 2Pb is formed but before the OLED layer 3 is formed, there may be a certain period of time (e.g., the element substrate is stored for at least one day or for several days) or the element substrate may be transported to a different plant. One method to prevent contamination of a surface of the organic flattening layer 2Pb during such a time period or to prevent adhesion of dust to the surface during the transportation may be, for example, forming a positive photoresist film that covers the organic flattening layer 2Pb. The photoresist film is preferably formed as follows. A photoresist solution is applied and then prebaking is performed (the solvent is volatilized to be removed; for example, baking is performed in a temperature range of about 90° C. or higher and about 110° C. or lower for about 5 minutes to about 30 minutes). The photoresist film is removed after the storage or the transportation but immediately before the formation of the OLED layer 3, so that a clean surface of the organic flattening layer 2Pb may be obtained. The photoresist film is preferably removed as follows. The photoresist film is entirely exposed to light and then is developed with no post-baking usually performed; or after the prebaking, the photoresist film is released by a release solution without being entirely exposed to light. A preferably usable material

to form the positive photoresist film is, for example, OFPR-800 produced by Tokyo Ohka Kogyo Co., Ltd., which is a positive photoresist.

(34) As described below with reference to FIG. 11B, a bank layer may be formed during the formation of the OLED element 3. The bank layer is formed after a lower electrode is formed but before an organic layer (organic EL layer) is formed. The bank layer is formed of an organic resin (e.g., polyimide), and therefore, easily absorbs moisture and contacts the organic layer. For these reasons, it is preferred to bake the bank layer to remove the moisture in the bank layer. After the organic flattening layer 2Pb is formed and also the lower electrode and the bank layer are formed, a positive photoresist film that protects a surface of the element substrate may be formed as described above. After the necessary storage and/or transportation, the photoresist film is removed as described above and the element substrate is baked immediately before the formation of the organic layer, so that the moisture contained in the organic flattening layer 2Pb and the bank layer is removed. In this manner, the baking step of removing the moisture contained in the organic flattening layer 2Pb and the baking step of removing the moisture contained in the bank layer may be performed as one step. Needless to say, the baking steps may be performed as follows. After the organic flattening layer 2Pb is protected with a positive photoresist film and the positive photoresist film is removed as described above, a baking step of removing the moisture contained in the organic flattening layer 2Pb is performed. After this, the lower electrode and the bank layer are formed, and then a baking step of removing the moisture contained in the bank layer is performed before the organic layer is formed.

(35) Now, with reference to FIG. 3A through FIG. 3C and FIG. 4A through FIG. 4D, the cross-sectional structure of the OLED display device 100A will be described in more detail.

(36) As shown in FIG. 3A, FIG. 3B, FIG. 4A and FIG. 4B, a TFE structure 10A includes the first inorganic barrier layer 12 formed on the OLED 3, the organic barrier layer 14 in contact with the first inorganic barrier layer 12 and the second inorganic barrier layer 16 in contact with the organic barrier layer 14. The first inorganic barrier layer 12 and the second inorganic barrier layer 16 are each, for example, an SiN.sub.x layer, and are each selectively formed by plasma CVD using a mask only in a predetermined region so as to cover the active region R1.

(37) The organic barrier layer 14 may be formed by, for example, the method described in Patent Document No. 2 or 3 mentioned above. For example, in a chamber, vapor of an organic material (e.g., acrylic monomer) or a mist-like organic material is supplied onto the element substrate maintained at a temperature lower than, or equal to, room temperature and is condensed on the element substrate. The organic material put into a liquid state is located locally, more specifically, at a border between a side surface of a protruding portion of, and a flat portion of, the first inorganic barrier layer 12 by a capillary action or a surface tension of the organic material. Then, the organic material is irradiated with, for example, ultraviolet rays to form a solid portion of the organic barrier layer (e.g., acrylic resin layer) 14 at the above-mentioned border in the vicinity of the protruding portion. The organic barrier layer 14 formed by this method does not substantially include the solid portion on the flat portion. Regarding the method for forming the organic barrier layer, the disclosures of Patent Documents Nos. 2 and 3 are incorporated herein by reference.

(38) Alternatively, the organic barrier layer 14 may be formed by adjusting an initial thickness of the resin layer to be formed by use of the film formation device 200 (e.g., to less than 100 nm) and/or by performing ashing on the resin layer once formed. As described below in detail, the ashing may be performed by plasma ashing using, for example, at least one type of gas among N.sub.2O, O.sub.2 and O.sub.3.

(39) FIG. 3A is a cross-sectional view taken along line 3A-3A' in FIG. 2, and shows a portion including a particle P. The particle P is a microscopic dust particle generated during the production of the OLED display device, and is, for example, a microscopic piece of broken glass, a metal particle or an organic particle. Such a particle is generated especially easily in the case where mask vapor deposition is used.

(40) As shown in FIG. 3A, the organic barrier layer (solid portion) **14** may be formed only in the vicinity of the particle P. A reason for this is that the acrylic monomer supplied after the first inorganic barrier layer **12** is formed is condensed and present locally, more specifically, in the vicinity of a surface of a first inorganic barrier layer **12a** on the particle P (the surface has a tapering angle θ of 90 degrees or larger). The opening (non-solid portion) of the organic barrier layer **14** is on the flat portion of the first inorganic barrier layer **12**.

(41) In the case where the particle (having a diameter of, for example, 1 μm or longer) P is present, a crack (defect) **12c** may be formed in the first inorganic barrier layer **12**. This is considered to be caused by impingement of the SiN.sub.x layer **12a** growing from a surface of the particle P and an SiN.sub.x layer **12b** growing from a flat portion of a surface of the OLED **3**. In the case where such a crack **12c** is present, the level of barrier property of the TFE structure **10A** is decreased.

(42) In the TFE structure **10A** in the OLED display device **100A**, as shown in FIG. 3A, the organic barrier layer **14** is formed to fill the crack **12c** of the first inorganic barrier layer **12**, and a surface of the organic barrier layer **14** couples the surface of the first inorganic barrier layer **12a** on the particle P and a surface of the first inorganic barrier layer **12b** on the flat portion of the OLED **3** to each other continuously and smoothly. Therefore, the second inorganic barrier layer **16**, which is formed on the first inorganic barrier layer **12a** on the particle P and also on the organic barrier layer **14**, is a fine film with no defect. As can be seen, even if the particle P is present, the organic barrier layer **14** may keep high the level of barrier property of the TFE structure **10A**.

(43) Now, with reference to FIG. 3B and FIG. 4A through FIG. 4D, a cross-sectional structure on the lead wires **32** and the terminals **34** will be described.

(44) As shown in FIG. 3B, the lead wires **32** and the terminals **34** are integrally formed on the substrate **1**. The inorganic protective layer **2Pa** is formed on the lead wires **32** so as to expose the terminals **34**. The organic flattening layer **2Pb** is formed on the inorganic protective layer **2Pa**, and the OLED layer **3** is formed on the organic flattening layer **2Pb**. The TFE structure **10A** is formed so as to cover the OLED layer **3** and the organic flattening layer **2Pb**. The OLED layer **3** and the organic flattening layer **2Pb** are enclosed by the joint portion where the inorganic protective layer **2Pa** and the first inorganic barrier layer **12** are in direct contact with each other. The organic barrier layer (solid portion) **14** between the first inorganic barrier layer **12** and the second inorganic barrier layer **16** in the TFE structure **10A** is formed only in the vicinity of a protruding portion such as the particle or the like, and thus is not shown here. The organic barrier layer (solid portion) **14** is enclosed by the inorganic barrier layer joint portion, where the first inorganic barrier layer **12** and the second inorganic barrier layer **16** are in direct contact with each other.

(45) As shown in FIG. 4A, in a region closer to the active region R1 (cross-section taken along line 4A-4A' in FIG. 2), the inorganic protective layer **2Pa**, the organic flattening layer **2Pb** and the TFE structure **10A** are formed on the lead wires **32**.

(46) As shown in FIG. 4B, in a cross-section taken along line 4B-4B' in FIG. 2, the inorganic protective layer **2Pa** and the first inorganic barrier layer **12** are in direct contact with each other, and the organic flattening layer **2Pb** is enclosed by the joint portion where the inorganic protective layer **2Pa** and the first inorganic barrier layer **12** are in direct contact with each other (see FIG. 2 and FIG. 3B).

(47) As shown in FIG. 4C, in a region closer to the terminals **34**, only the inorganic protective layer **2Pa** is formed on the lead wires **32**.

(48) As shown in FIG. 4D, the terminals **34** are exposed also from the inorganic protective layer **2Pa** and are used for electrical connection with an external circuit (e.g., FPCs (Flexible Printed Circuit)).

(49) A region including the portions shown in FIG. 4B through FIG. 4D is not covered with the organic flattening layer **2Pb**. Therefore, during the formation of the organic barrier layer **14** of the TFE structure **10A**, an organic barrier layer (solid portion) may be formed in this region. For example, if a side surface of a cross-section, of the lead wires **32**, that is parallel to the line width

direction thereof has a tapering angle $\square$ of 90 degrees or larger, an organic barrier layer may be formed along the side surface of each of the lead wires **32**. However, as shown in FIG. **4B** through FIG. **4D**, in the OLED display device **100A** according to this embodiment, at least in this region, the tapering angle $\square$ of the side surface of the cross-section of each of the lead wires **32** and each of the terminals **34** is smaller than 90 degrees, and thus the photocurable resin is not located locally. Therefore, the organic barrier layer (solid portion) is not formed along the side surface of any of the lead wires **32** or any of the terminals **34**.

(50) Now, with reference to FIG. **3C**, the tapering angle $\square$ of the side surface of each of the layers will be described. FIG. **3C** is a cross-sectional view showing the tapering angle $\square$ of the side surface of each of the layers, and corresponds to, for example, the cross-section of FIG. **4B**. As shown in FIG. **3C**, the tapering angle $\square$ of the side surface of the cross-section, of the lead wire **32**, that is parallel to the width direction thereof will be represented as $\square(32)$. The tapering angle $\square$ of the side surface of each of the other layers will be represented in a similar manner with $\square$ (reference sign of the component).

(51) With such representations, the tapering angles $\square$ of the side surface of the inorganic protective layer **2Pa** formed on the lead wire **32**, the side surface of the first inorganic barrier layer **12** of the TFE structure **10A**, formed on the inorganic protective layer **2Pa**, and the side surface of the second inorganic barrier layer **16** of the TFE structure **10A** formed on the inorganic protective layer **2Pa** fulfill the relationship of $\square(32) \geq \square(2Pa) \geq \square(12) \geq \square(16)$. Therefore, if the tapering angle $\square(32)$ of the side surface of the lead wire **32** is smaller than 90 degrees, the tapering angle $\square(2Pa)$ of the side surface of the inorganic protective layer **2Pa** and the tapering angle $\square(12)$ of the side surface of the first inorganic barrier layer **12** are each also smaller than 90 degrees.

(52) Assuming that the method for forming the organic barrier layer described in Patent Document No. 2 or 3 is used in the case where the tapering angle $\square$ of the side surface is 90 degrees or larger, vapor of an organic material (e.g., acrylic monomer) or a mist-like organic material is condensed, and thus an organic barrier layer (solid portion) is formed, along a border between the side surface and the flat surface (making an angle of 90 degrees or smaller). When this occurs, for example, the organic barrier layer (solid portion) formed along the lead wire acts as a route that guides water vapor in the air into the active region.

(53) FIG. **5A** is a schematic cross-sectional view of an OLED display device **100B1** according to a comparative example, and corresponds to FIG. **4B**. If, as shown in FIG. **5A**, a tapering angle $\square(32B1)$ of a side surface of each of lead wires **32B1** and a tapering angle $\square(12B1)$ of a side surface of a first inorganic barrier layer **12B1** are each 90 degrees or larger, an organic barrier layer (solid portion) **14B1** is formed between the first inorganic barrier layer **12B1** and a second inorganic barrier layer **16B1** of a TFE structure **10B1** along the side surface of the first inorganic barrier layer **12B1**. The OLED display device **100B1** may be obtained by, for example, changing the OLED display device **100** according to an embodiment as follows. The inorganic protective layer **2Pa** is omitted, and the tapering angle $\square(32)$ of the side surface of the lead wire **32** and the tapering angle $\square(12)$ of the side surface of the first inorganic barrier layer **12** are changed to 90 degrees or larger.

(54) FIG. **5B** is a schematic cross-sectional view of an OLED display device **100B2** according to a comparative example, and corresponds to FIG. **4B**. If, as shown in FIG. **5B**, a tapering angle $\square(32B2)$ of a side surface of each of lead wires **32B2**, a tapering angle $\square(2PaB2)$ of a side surface of an inorganic protective layer **2PaB2** and a tapering angle $\square(12B2)$ of a side surface of a first inorganic barrier layer **12B2** are each 90 degrees or larger, an organic barrier layer (solid portion) **14B2** is formed between the first inorganic barrier layer **12B2** and a second inorganic barrier layer **16B2** of a TFE structure **10B2** along the side surface of the first inorganic barrier layer **12B2**. The OLED display device **100B2** may be obtained by, for example, changing the OLED display device **100A** according to an embodiment as follows. The tapering angle $\square(32)$ of the side surface of the lead wire **32** and the tapering angle $\square(12)$ of the side surface of the first inorganic barrier layer **12**

are changed to 90 degrees or larger.

(55) Unlike the OLED display device **100B1**, the OLED display device **100B2** includes the inorganic protective layer **2PaB2**. Therefore, the tapering angle $\square(12B2)$ of the side surface of the first inorganic barrier layer **12B2** is likely to be smaller than the tapering angle $\square(12B1)$ of the side surface of the first inorganic barrier layer **12B1** of the OLED display device **100B1**.

(56) In the OLED display device **100A** in an embodiment according to the present invention shown in FIG. 4B through FIG. 4D, the tapering angles $\square(32)$, $\square(2Pa)$ and $\square(12)$ of the side surfaces of the lead wire **32**, the inorganic protective layer **2Pa** and the first inorganic barrier layer **12** are all smaller than 90 degrees. Thus, the organic barrier layer **14** is not formed along these side surfaces. Therefore, moisture in the air does not reach the inside of the active region **R1** via the organic barrier layer (solid portion) **14**, and thus the OLED display device **100A** may have a high level of moisture-resistant reliability. In this example, the tapering angles $\square(32)$, $\square(2Pa)$ and $\square(12)$ are all smaller than 90 degrees. The present invention is not limited to this. It is sufficient that the tapering angle $\square(12)$ of the side surface, of the first inorganic barrier layer **12**, which forms a surface immediately below the organic barrier layer **14** has a tapering angle smaller than 90 degrees. With such a structure, the stack structure shown in FIG. 4B (the portion where the inorganic protective layer **2Pa** and the first inorganic barrier layer **12** are in direct contact with each other (the organic flattening layer **2Pb** is not present), and the portion where the first inorganic barrier layer **12** and the second inorganic barrier layer **16** are in direct contact with each other (the organic barrier layer **14** is not present)) is formed. Therefore, the situation may be suppressed or prevented where moisture in the air enters the inside of the active region **R1** via the organic flattening layer **2Pb** or the organic barrier layer **14**. In addition, the provision of the inorganic protective layer **2Pa** may decrease the tapering angle $\square(12)$ of the first inorganic barrier layer **12**. Therefore, even if the tapering angle $\square(32)$ of the lead wire **32** is relatively large (e.g., 90 degrees), the tapering angle $\square(12)$ of the first inorganic barrier layer **12** may be made smaller than 90 degrees. Namely, the tapering angle $\square(32)$ of the lead wire **32** may be made 90 degrees or closer to 90 degrees. This provides an advantage that L/S of the lead wire **32** may be made smaller.

(57) In the case where the tapering angle $\square$ of the side surface is in the range of 70 degrees or larger and smaller than 90 degrees, the organic barrier layer (solid portion) **14** may be formed along the side surface. Needless to say, the resin present locally, namely, along the inclining side surface, may be removed by ashing. However, the ashing is time-consuming. For example, the ashing needs to be performed for a long time even after the resin formed on the flat surface is removed. Or, there may be a problem that as a result of the organic barrier layer (solid portion) formed in the vicinity of the particle **P** being excessively asked (removed), the effect of the formation of the organic barrier layer is not fully provided. In order to suppress or prevent this problem, the tapering angle $\theta(12)$ of the first inorganic barrier layer **12** is preferably smaller than 70 degrees, and more preferably smaller than 60 degrees.

(58) Now, with reference to FIG. 6 through FIG. 8, a structure of the OLED display device **100C** according to a comparative example will be described. FIG. 6 is a schematic plan view of the OLED display device **100C**. FIG. 7A and FIG. 7B are each a schematic cross-sectional view of the OLED display device **100C**. FIG. 7A is a cross-sectional view taken along line 7A-7A' in FIG. 6, and FIG. 7B is a cross-sectional view taken along line 7B-7B' in FIG. 6. FIG. 8A through FIG. 8C are each a schematic cross-sectional view of the OLED display device **100C**. FIG. 8A is a cross-sectional view taken along line 8A-8A' in FIG. 6. FIG. 8B is a cross-sectional view taken along line 8B-8B' in FIG. 6. FIG. 8C is a cross-sectional view taken along line 8C-8C' in FIG. 6.

(59) Unlike in the OLED display device **100A** according to an embodiment, in the OLED display device **100C**, the inorganic protective layer **2Pa** is not included, and the organic flattening layer **2Pbc** is extended to a region that is not covered with a TFE structure **10C**. Components substantially the same as those of the OLED display device **100A** bear the identical reference signs thereto and descriptions thereof may be omitted.

(60) As is clear from, for example, FIG. 6, FIG. 7B and FIG. 8B, the organic flattening layer 2Pbc is partially exposed to the air (ambient atmosphere). In this case, the organic flattening layer 2Pbc absorbs moisture through the portion exposed to the air and acts as a route that guides water vapor in the air into the active region R1. By contrast, in the OLED display device 100A according to an embodiment, as shown in FIG. 3B and FIG. 4B, the organic flattening layer 2Pb is enclosed, together with the OLED layer 3, by the joint portion where the inorganic protective layer 2Pa and the first inorganic barrier layer 12 are in direct contact with each other. Therefore, the above-described problem of the OLED display device 100C according to a comparative example may be solved.

(61) Now, with reference to FIG. 9 and FIG. 10, examples of TFT usable for the OLED display device 100A, and examples of lead wires and terminals formed by use of a gate metal layer and a source metal layer used to form the TFT, will be described. The structures of the TFT, the lead wires and the terminals described below may be used for the OLED display device 100A described above.

(62) For a medium- or small-sized high-definition OLED display device, a low temperature polycrystalline silicon (hereinafter, referred to simply as “LTPS”) TFT or an oxide TFT (e.g., four-component-based (In—Ga—Zn—O-based) oxide TFT containing In (indium), Ga (gallium), Zn (zinc) and O (oxygen)) having a high mobility is preferably used. Structures of, and methods for producing, the LTPS-TFT and the In—Ga—Zn—O-based TFT are well known and will be described below merely briefly.

(63) FIG. 9A is a schematic cross-sectional view of an LTPS-TFT 2.sub.pT. The TFT 2.sub.pT may be included in the circuit 2 of the OLED display device 100A. The LTPS-TFT 2.sub.pT is a top gate-type TFT.

(64) The TFT 2.sub.pT is formed on a base coat 2.sub.pp on the substrate (e.g., polyimide film) 1. Although not described above, it is preferred that a base coat formed of an inorganic insulating material is formed on the substrate 1.

(65) The TFT 2.sub.pT includes a polycrystalline silicon layer 2.sub.pse formed on the base coat 2.sub.pp, a gate insulating layer 2.sub.pgi formed on the polycrystalline silicon layer 2.sub.pse, a gate electrode 2.sub.pg formed on the gate insulating layer 2.sub.pgi, an interlayer insulating layer 2.sub.pi formed on the gate electrode 2.sub.pg, and a source electrode 2.sub.pss and a drain electrode 2.sub.psd formed on the interlayer insulating layer 2.sub.pi. The source electrode 2.sub.pss and the drain electrode 2.sub.psd are respectively connected with a source region and a drain region of the polycrystalline silicon layer 2.sub.pse in contact holes formed in the interlayer insulating layer 2.sub.pi and the gate insulating layer 2.sub.pgi.

(66) The gate electrode 2.sub.pg is included in a gate metal layer including the gate bus lines, and the source electrode 2.sub.pss and the drain electrode 2.sub.psd are included in a source metal layer including the source bus lines. The gate metal layer and the source metal layer are used to form lead wires and terminals (described below with reference to FIG. 10).

(67) The TFT 2.sub.pT is formed, for example, as follows.

(68) As the substrate 1, for example, a polyimide film having a thickness of 15 μm is prepared.

(69) The base coat 2.sub.pp (SiO.sub.2 film: 250 nm/SiN.sub.x film: 50 nm/SiO.sub.2 film: 500 nm (top layer/middle layer/bottom layer)) and an a-Si film (40 nm) are formed by plasma CVD.

(70) The a-Si film is subjected to dehydrogenation (e.g., annealed at 450°C for 180 minutes).

(71) The a-Si film is polycrystalline-siliconized by excimer laser annealing (ELA).

(72) The a-Si film is patterned by a photolithography step to form an active layer (semiconductor island).

(73) A gate insulating film (SiO.sub.2 film: 50 nm) is formed by plasma CVD.

(74) A channel region of the active layer is doped with (B.sup.+).

(75) The gate metal layer (Mo: 250 nm) is formed by sputtering and patterned by a photolithography step (including a dry etching step) (to form the gate electrode 2.sub.pg, the gate

bus lines, and the like).

(76) The source region and the drain region of the active layer are doped with (P.sup.+).

(77) Activation annealing (e.g., annealing at 450°C for 45 minutes) is performed. As a result, the polycrystalline silicon layer 2.sub.pse is formed.

(78) An interlayer insulating film (e.g., SiO.sub.2 film: 300 nm/SiN.sub.x film: 300 nm (top layer/bottom layer)) is formed by plasma CVD.

(79) The contact holes are formed in the gate insulating film and the interlayer insulating film by dry etching. As a result, the interlayer insulating layer 2.sub.pi and the gate insulating layer 2.sub.pgi are formed.

(80) The source metal layer (Ti film: 100 nm/Al film: 300 nm/Ti film: 30 nm) is formed by sputtering and patterned by a photolithography step (including a dry etching step) (to form the source electrode 2.sub.pss, the drain electrode 2.sub.psd, the source bus lines, and the like).

(81) After this, the inorganic protective layer 2Pa (see FIG. 2 and FIG. 3) described above is formed by, for example, plasma CVD.

(82) FIG. 9B is a schematic cross-sectional view of an In—Ga—Zn-TFT 2.sub.oT. The TFT 2.sub.oT may be included in the circuit 2 of the OLED display device 100A. The TFT 2.sub.oT is a bottom gate-type TFT.

(83) The TFT 2.sub.oT is formed on a base coat 2.sub.op on the substrate 1 (e.g., polyimide film). The TFT 2.sub.oT includes a gate electrode 2.sub.og formed on the base coat 2.sub.op, a gate insulating layer 2.sub.ogi formed on the gate electrode 2.sub.og, an oxide semiconductor layer 2.sub.ose formed on the gate insulating layer 2.sub.ogi, and a source electrode 2.sub.oss and a drain electrode 2.sub.osd respectively formed on a source region and a drain region of the oxide semiconductor layer 2.sub.ose. The source electrode 2.sub.oss and the drain electrode 2.sub.osd are covered with an interlayer insulating layer 2.sub.oi.

(84) The gate electrode 2.sub.og is included in a gate metal layer including the gate bus lines, and the source electrode 2.sub.oss and the drain electrode 2.sub.osd are included in a source metal layer including the source bus lines. The gate metal layer and the source metal layer are used to form lead wires and terminals, and thus the TFT 2.sub.oT may have a structure described below with reference to FIG. 10.

(85) The TFT 2.sub.oT is formed, for example, as follows.

(86) As the substrate 1, for example, a polyimide film having a thickness of 15 μm is prepared.

(87) The base coat 2.sub.op (SiO.sub.2 film: 250 nm/SiN.sub.x film: 50 nm/SiO.sub.2 film: 500 nm (top layer/middle layer/bottom layer)) is formed by plasma CVD.

(88) The gate metal layer (Cu film: 300 nm/Ti film: 30 nm (top layer/bottom layer)) is formed by sputtering and patterned by a photolithography step (including a dry etching step) (to form the gate electrode 2.sub.og, the gate bus lines, and the like).

(89) A gate insulating film (SiO.sub.2 film: 30 nm/SiN.sub.x film: 350 nm (top layer/bottom layer)) is formed by plasma CVD.

(90) An oxide semiconductor film (In—Ga—Z—O-based semiconductor film: 100 nm) is formed by sputtering and patterned by a photolithography step (including a wet etching step) to form an active layer (semiconductor island).

(91) The source metal layer (Ti film: 100 nm/Al film: 300 nm/Ti film: 30 nm (top layer/middle layer/bottom layer)) is formed by sputtering and patterned by a photolithography step (including a dry etching step) (to form the source electrode 2.sub.oss, the drain electrode 2.sub.osd, the source bus lines, and the like).

(92) Activation annealing (e.g., annealing at 300°C for 120 minutes) is performed. As a result, the oxide semiconductor layer 2.sub.ose is formed.

(93) Then, the interlayer insulating layer 2.sub.oi (e.g., SiN.sub.x film: 300 nm/SiO.sub.2 film: 300 nm (top layer/bottom layer)) is formed by plasma CVD as a protective film. The interlayer insulating layer 2.sub.oi may also act as the inorganic protective layer 2Pa (see FIG. 2 and FIG. 3)

described above. Needless to say, the inorganic protective layer 2Pa may further be formed on the interlayer insulating layer 2.sub.oi.

(94) Now, with reference to FIG. 10A through FIG. 10C, a structure of another OLED display device according to an embodiment will be described. The circuit (backplane) 2 of this OLED display device includes the TFT 2.sub.pT shown in FIG. 9A or the TFT 2.sub.oT shown in FIG. 9B. The gate metal layer and the source metal layer used to form the TFT 2.sub.pT or the TFT 2.sub.oT are used to form a lead wire 32A and a terminal 34A. FIG. 10A through FIG. 10C respectively correspond to FIG. 4B through FIG. 4D. Components corresponding to those in FIG. 4B through FIG. 4D will be represented by the identical reference signs thereto provided with letter “A” at the end. A base coat 2p in FIG. 10 corresponds to the base coat 2.sub.pp in FIG. 9A and the base coat 2.sub.op in FIG. 9B. A gate insulating layer 2gi in FIG. 10 corresponds to the gate insulating layer 2.sub.pgi in FIG. 9A and the gate insulating layer 2.sub.ogi in FIG. 9B. An interlayer insulating layer 2i in FIG. 10 corresponds to the interlayer insulating layer 2.sub.pi in FIG. 9A and the interlayer insulating layer 2.sub.oi in FIG. 9B.

(95) As shown in FIG. 10A through FIG. 10C, a gate metal layer 2g and a source metal layer 2s are formed on the base coat 2p, which is formed on the substrate 1. Although not shown in FIG. 3 or FIG. 4, it is preferred that the base coat 2p formed of an inorganic insulating material is formed on the substrate 1.

(96) As shown in FIG. 10p, through FIG. 10C, the lead wire 32A and the terminal 34A are each formed as a stack body of the gate metal layer 2g and the source metal layer 2s. A portion of the lead wire 32A and a portion of the terminal 34A that are formed of the gate metal layer 2g have, for example, the same cross-sectional shape as that of the gate bus lines. A portion of the lead wire 32A and a portion of the terminal 34A that are formed of the source metal layer 2s have, for example, the same cross-sectional shape as that of the source bus lines. In the case of, for example, a 5.7-type display device of 500 ppi, the portion formed of the gate metal layer 2g has a line width of, for example, 10 μm , and a distance between two adjacent such lines is 16 μm ($L/S=10/16$). The portion formed of the source metal layer 2s has a line width of, for example, 16 μm , and a distance between two adjacent such lines is 10 μm ($L/S=16/10$). These portions each have a tapering angle $\square$ smaller than degrees, preferably smaller than 70 degrees, and more preferably degrees or smaller. The tapering angle of portions formed below the organic flattening layer Pb may be 90 degrees or larger.

(97) Now, with reference to FIG. 11, a structure of, and a method for producing, another OLED display device according to another embodiment of the present invention will be described. The OLED display device shown in FIG. 11 includes an LTPS-TFT 2.sub.pT shown in FIG. 9A. Needless to say, the In—Ga—Zn—O-based TFT 2.sub.oT shown in FIG. 9B may be used or any other known TFT may be used instead of the LTPS-TFT 2.sub.pT. FIG. 11A and FIG. 11B also show a common wire 2.sub.pc and a contact portion 2.sub.pcV to be electrically connected with an upper electrode (common electrode) 46 of the OLED.

(98) Unlike in the above-described embodiments, the OLED display device 100D according to this embodiment further includes the second inorganic protective layer 2Pa2 formed on the organic flattening layer 2Pb. A surface of the second inorganic protective layer 2Pa2 has an arithmetic average roughness Ra of 50 nm or less.

(99) The lower electrode and the organic layer (also referred to as an “organic EL layer”; includes at least an organic light emitting layer) of the OLED 3 are required to have a high flatness. If the flatness is low, for example, the OLED 3 has a low light emitting efficiency. The organic flattening layer 2Pb is formed of a liquid-state organic resin material (may contain a solvent) applied to the surface of the element substrate (the element substrate in the middle of production is also referred to as an “element substrate” for the sake of simplicity), and therefore, may absorb the recesses and protrusions (steps) of an underlying layer for the organic flattening layer 2Pb and form a flat surface. However, the surface of the organic flattening film formed of the liquid-state resin material

has an arithmetic average roughness Ra (may be represented as “waviness”) of greater than 100 nm and about 300 nm or less (see FIG. 11A). If the OLED 3 is formed on the organic flattening layer 2Pb having such a surface roughness, the light emitting efficiency may not be sufficiently improved. There is a problem that, for example, even if a micro-cavity structure is adopted, the effect thereof may not be sufficiently expressed.

(100) Under such circumstances, according to an embodiment of the present invention, the second inorganic protective film is formed on the organic resin film formed as described above, and a surface of the second inorganic protective film is subjected to chemical mechanical polishing (hereinafter, referred to as “CMP”). The CMP allows the arithmetic average roughness Ra of the second inorganic protective film to be controlled to be 50 nm or less, preferably nm or less. A smaller arithmetic average roughness Ra is more preferred, and there is no limitation on the lowest value thereof. The arithmetic average roughness Ra may be 10 nm or greater, or nm or greater, in consideration of the relationship between the cost or time required for the CMP and the effect thereof.

(101) The surface roughness may be measured by use of, for example, a confocal laser scanning microscope or an atomic force microscope (AFM). It is preferred that the range of measurement encompasses the center of the pixel and the vicinity thereof, and the reference length is appropriately set in accordance with the surface roughness.

(102) In this specification, in order to distinguish flattening by the organic flattening layer 2Pb and flattening by the CMP from each other, the flattening by the CMP may be referred to as “smoothing”.

(103) FIG. 11A schematically shows a state immediately after the first inorganic protective film 2Pa1 to be the first inorganic protective layer 2Pa1, the organic flattening film 2Pb to be the organic flattening layer 2Pb, and the second inorganic protective film 2Pa2 to be the second inorganic protective layer 2Pa2 are formed on the driving circuit layer 2. The first inorganic protective film 2Pa1 and the second inorganic protective film 2Pa2 are each selectively formed in a desired region by use of a metal mask. The organic flattening film 2Pb is selectively formed in a desired region of the element substrate 20 by application of a material thereof, or is once formed on the entire surface of the element substrate 20 and then an unnecessary portion is removed. The first inorganic protective film 2Pa1 and the second inorganic protective film 2Pa2, which do not include an opening such as a contact hole or the like, are different from the first inorganic protective layer 2Pa1 and the second inorganic protective layer 2Pa2, but are represented by the identical reference signs thereto for the sake of simplicity. Similarly, the organic flattening film 2Pb, which does not include an opening such as a contact hole or the like, is different from the organic flattening layer 2Pb, but is represented by the identical reference sign thereto for the sake of simplicity. The first inorganic protective layer 2Pa1 and the second inorganic protective layer 2Pa2 are formed in substantially the same manner as the inorganic protective layer 2Pa described above.

(104) A surface 2Pb_Sa of the organic flattening film 2Pb has an arithmetic average roughness Ra of greater than 100 nm and about 300 nm or less. In the case where the organic flattening film 2Pb is formed of a non-photosensitive resin, a surface having a smaller arithmetic average roughness may be obtained because of an effect provided by adding a leveling agent or the like than in the case where the organic flattening film 2Pb is formed of a photosensitive resin. Nonetheless, the arithmetic average roughness Ra of the surface exceeds 50 nm.

(105) The surface roughness of a surface 2Pa2_Sa of the second inorganic protective film 2Pa2 formed on the surface 2Pb_Sa of the organic flattening film 2Pb described above reflects the surface roughness of the surface 2Pb_Sa of the organic flattening film 2Pb, and exceeds 50 nm.

(106) Under such circumstances, in the OLED display device 100D according to an embodiment of the present invention, as shown in FIG. 11B, a surface 2Pa2_Sb of the second inorganic protective film 2Pa2 is smoothed, and has an arithmetic average roughness Ra controlled to be 50 nm or less.

(107) When being formed, the second inorganic protective film 2Pa2 has a thickness (FIG. 11A)

that is at least about three times the arithmetic average roughness Ra of the surface 2Pb_Sa of the organic flattening film 2Pb, and has the surface thereof treated with CMP. The resultant surface 2Pa2_Sb has an arithmetic average roughness Ra of 50 nm or less. It is preferred that the arithmetic average roughness Ra of the surface 2Pa2_Sb is 30 nm or less. A smaller arithmetic average roughness Ra is more preferred, and there is no limitation on the lowest value thereof. The arithmetic average roughness Ra may be 10 nm or greater, or may be 20 nm or greater, in consideration of the relationship between the cost or time required for the CMP and the effect thereof.

(108) In the case where, for example, the second inorganic protective film 2Pa2 is formed as a single layer of SiN.sub.x film by plasma CVD, the SiN.sub.x film may be formed at a film formation rate that does not exceed, for example, 10 nm/s. In this case, the arithmetic average roughness Ra of the surface 2Pb_Sa of the organic flattening film 2Pb, which is an underlying layer for the second inorganic protective film 2Pa2, is reflected almost as it is by the surface 2Pa2_Sa of the second inorganic protective film 2Pa2. Then, the second inorganic protective film 2Pa2 is treated with the CMP, and as a result, the surface 2Pa2_Sb of the second inorganic protective film 2Pa2 is smoothed to have an arithmetic average roughness Ra of 50 nm or less.

(109) The CMP is performed by use of, for example, a slurry containing a neutral ceria (CeO.sub.2)-based polishing agent or a slurry containing a fumed silica-based or colloidal silica-based polishing agent. Alternatively, a slurry containing alumina (Al.sub.2O.sub.3) powder may be used. As a chemical solution, a neutral aqueous solution or an alcohol is usable. Alternatively, in the case where the substrate 1 is a glass substrate, an aqueous solution of KOH is usable as the chemical solution. In the case where the substrate 1 is a polyimide film, a neutral chemical solution is preferred. A preferred load of a polishing pad is 50 g/cm.sup.2 or greater and 150 g/cm.sup.2 or less. A preferred rotation rate is 20 rpm or greater and 40 rpm or less. Polishing requires about 30 seconds to about 150 seconds.

(110) In an OLED display device according to an embodiment of the present invention, the level of smoothing required to suppress the decrease in the light emitting efficiency is lower by one digit (the value of the arithmetic average roughness Ra is about ten times larger) than the level of smoothing required for a photographic process for a VLSI or a surface of an optical lens. Therefore, any of known polishing agents in a wide range generally usable for CMP is usable.

(111) Smoothing the surface of the second inorganic protective layer 2Pa2 allows the lower electrode, the organic layer and the upper electrode of the OLED 3 formed on the second inorganic protective layer 2Pa2 to be smooth. This may improve the light emitting efficiency of the OLED display device.

(112) Next, the first inorganic protective film 2Pa1, the organic flattening film 2Pb and the second inorganic protective film 2Pa2 each formed in a desired region of the element substrate 20 are patterned in one step to form openings such as contact holes CH1 and CH2 or the like. The patterning may be performed by, for example, dry etching by use of mixed gas of O.sub.2 and CF.sub.4. Therefore, the resin forming the organic flattening film 2Pb does not need to be photosensitive. As shown in FIG. 11C, there is a portion where only the first inorganic protective layer 2Pa1 is left in order to form a portion where the first inorganic protective layer 2Pa1 and the first inorganic barrier layer (not shown) in the TFE structure 10D are in direct contact with each other. Such a structure may be formed by use of, for example, a gray-scale mask.

(113) Next, a lower electrode 42 and an upper electrode contact portion 43 are formed. The lower electrode 42 is electrically connected with the drain electrode 2.sub.psd in the contact hole CH1.

(114) Next, a bank layer 48 defining each of the plurality of pixels Pix is formed. The bank layer 48 includes an opening corresponding to each of the pixels Pix. A side surface of the opening has an inclining surface including a forward tapering side surface portion. The inclining surface of the bank layer 48 encloses the corresponding pixel. The bank layer 48 is formed of, for example, a photosensitive resin (e.g., polyimide or acrylic resin). The bank layer 48 has a thickness of, for

example, 1 μ m or greater and 2 μ m or less. The inclining surface of the bank layer **48** is inclined at an inclination angle of 60 degrees or smaller. If the inclination angle α of the inclining surface of the bank layer **48** is larger than 60 degrees, a defect may be caused in layers located on the bank layer **48**.

(115) As described above, it is preferred to bake the resultant layers in order to remove moisture contained in the bank layer **48** and moisture contained in the organic flattening layer **2Pb**. The baking temperature is preferably, for example, 100° C. or higher (e.g., for 1 hours or longer), and more preferably 250° C. or higher (e.g., for 15 minutes or longer). The atmosphere may be an atmospheric pressure. Baking in a low pressure atmosphere may shorten the baking time. It is preferred that the bank layer **48** and the organic flattening layer **2Pb** are formed of a resin material having a high thermal resistance, for example, polyimide, in order to prevent deterioration by heat in the baking step.

(116) After this, it may occur that the production of the element substrate is suspended and the element substrate is stored or transported. In such a case, a positive photoresist film may be formed to cover the surface of the element substrate including the layers up to the bank layer **48** as described above. After the storage or the transportation, the photoresist film is removed immediately before the organic layer **44** is formed, so that a clean surface may be provided. Immediately before the organic layer **44** is formed, the layers are baked to remove the moisture contained in the organic flattening layer **2Pb** and the bank layer **48**.

(117) Next, the organic layer **44** is formed on the lower electrode **42**. The organic layer **44** is formed by, for example, vapor deposition. The upper electrode **46** is formed on the organic layer **44**. The upper electrode **46** is in contact with the upper electrode contact portion **43**, and is electrically connected with the common wire 2.sub.pc via the contact portion 2.sub.pcV.

(118) The lower electrode **42** and the upper electrode **46** respectively act as, for example, an anode and a cathode. The upper electrode **46** is a common electrode formed for the entirety of the pixels in the active region. The lower electrode (pixel electrode) **42** is formed for each of the pixels. In the structure in which the bank layer **48** is present between the lower electrode **42** and the organic layer **44**, no holes are implanted from the lower electrode **42** into the organic layer **44**. Therefore, the region where the bank layer **48** is present does not act as a pixel Pix. For this reason, the bank layer **48** defines an outer perimeter of each of the pixels Pix. The bank layer **48** may be referred to as a "PDL (Pixel Defining Layer)".

(119) After the OLED **3** is formed as described above, the TFE structure **10D** is formed. The TFE structure **10D** has substantially the same structure as that of the TFE structure **10A**, described above. Namely, the TFE structure **10D** includes the first inorganic barrier layer (e.g., SiN.sub.x layer) **12**, the organic barrier layer **14** and the second inorganic barrier layer (e.g., SiN.sub.x layer) **16**, provided in this order with the first inorganic barrier layer **12** being closest to the OLED **3**. As shown in FIG. **11C**, the first inorganic barrier layer **12** is in contact with the top surface and an inclining side surface of the second inorganic protective layer **2Pa2**, an inclining side surface of the organic flattening layer **2Pb** and a top surface of the first inorganic protective layer **2Pa1**. Namely, the OLED display device **100D** has substantially the same structure as that of the OLED display device **100A** shown in FIG. **3B** at an end thereof, and thus has a superb moisture-resistance reliability.

(120) Now, with reference to FIG. **12A** and FIG. **12B**, the film formation device **200** usable to form an organic barrier layer, and a film formation method using the same will be described. FIG. **12A** and FIG. **12B** schematically show a structure of the film formation device **200**. FIG. **12A** shows a state of the film formation device **200** in a step of, in a chamber having vapor of a photocurable resin or a mist-like photocurable resin located therein, condensing the photocurable resin on the first inorganic barrier layer. FIG. **12B** shows a state of the film formation device **200** in a step of irradiating the photocurable resin with light to which the photocurable resin is sensitive and thus curing the photocurable resin.

(121) The film formation device **200** includes a chamber **210** and a partition wall **234** dividing an inner space of the chamber **210** into two spaces. In one of the spaces, in the chamber **210**, demarcated by the partition wall **234**, a stage **212** and a shower plate **220** are located. In the other space demarcated by the partition wall **234**, an ultraviolet irradiation device **230** is located. The inner space of the chamber **210** is controlled to have a predetermined pressure (vacuum degree) and a predetermined temperature. The stage **212** has a top surface that receives the element substrate **20** including the plurality of OLEDs **3**, on which the first inorganic barrier layer is formed. The top surface may be cooled down to, for example, -20°C .

(122) The shower plate **220** is located to have a gap **224** between the shower plate **220** and the partition wall **234**. The shower plate **220** includes a plurality of through-holes **222**. The gap **224** may have a size of, for example, 100 mm or longer and 1000 mm or shorter in a vertical direction. An acrylic monomer (vapor or mist-like) supplied to the gap **224** is supplied, via the plurality of through-holes **222** of the shower plate **220**, to one of the spaces of the chamber **210** in which the stage **212** is located. When necessary, the acrylic monomer is heated. Vapor of an acrylic monomer or a mist-like acrylic monomer **26p** is attached to, or contacts, the first inorganic barrier layer on the element substrate **20**. An acrylic monomer **26** is supplied from a container **202** into the chamber **210** at a predetermined flow rate. The container **202** is supplied with the acrylic monomer **26** via a pipe **206** and is also supplied with nitrogen gas from a pipe **204**. The flow rate of the acrylic monomer supplied to the container **202** is controlled by a mass flow controller **208**. A material supply device includes the shower plate **220**, the container **202**, the pipes **204** and **206**, the mass flow controller **208** and the like.

(123) The ultraviolet irradiation device **230** includes an ultraviolet light source and optional optical elements. The ultraviolet light source may be, for example, an ultraviolet lamp (e.g., mercury lamp (encompassing a high-pressure lamp and a super-high pressure lamp), a mercury-xenon lamp or a metal halide lamp). The optical elements are, for example, a reflective mirror, a prism, a lens, and a diffraction element.

(124) The ultraviolet irradiation device **230** emits light having a predetermined wavelength and a predetermined intensity toward the top surface of the stage **212** when being located at a predetermined position. It is preferred that the partition wall **234** and the shower plate **220** are formed of a material having a high transmittance to ultraviolet rays, for example, quartz.

(125) The organic barrier layer **14** may be formed, for example, as follows by use of the film formation device **200**. In this example, an acrylic monomer is used as the photocurable resin.

(126) The acrylic monomer **26p** is supplied into the chamber **210**. The element substrate **20** has been cooled to, for example, -15°C on the stage **212**. The acrylic monomer **26p** is condensed on the first inorganic barrier layer **12** on the element substrate **20**. The conditions in this step may be controlled such that the liquid-state acrylic monomer is present locally, more specifically, only in the vicinity of the protruding portion of the first inorganic barrier layer **12**. Alternatively, the conditions may be controlled such that the acrylic monomer condensed on the first inorganic barrier layer **12** forms a liquid film.

(127) The viscosity and/or the surface tension of the photocurable resin in the liquid state may be adjusted to control the thickness of the liquid film or the shape of the portion of the liquid film that is to be in contact with the protruding portion of the first inorganic barrier layer **12** (namely, the shape of the recessed portion). For example, the viscosity and the surface tension depend on the temperature, and therefore, the temperature of the element substrate may be adjusted to control the viscosity and the surface tension. For example, the size of the solid portion that is present on the flat portion may be controlled by the shape of a portion, of the liquid film, that is to be in contact with the protruding portion of the first inorganic barrier layer **12** (namely, the shape of the recessed portion) and by the conditions of asking to be performed in a later step.

(128) Next, the ultraviolet irradiation device **230** is used to, typically, irradiate the entirety of a top surface of the element substrate **20** with ultraviolet rays **232** and thus to cure the acrylic monomer

on the first inorganic barrier layer **12**. As the ultraviolet light source, for example, a high-pressure mercury lamp that provides light having a main peak wavelength of 365 nm is used. The ultraviolet rays are directed at an intensity of, for example, 12 mW/cm² for about 10 seconds.

(129) The organic barrier layer **14** formed of an acrylic resin is formed in this manner. The tact time of the step of forming the organic barrier layer **14** is, for example, shorter than about 30 seconds. Thus, the mass-productivity is very high.

(130) Alternatively, after the photocurable resin in a liquid film state is cured, ashing may be performed to form the organic barrier layer **14** only in the vicinity of the protruding portion. Also in the case where the organic barrier layer **14** is formed by curing the photocurable resin present locally, the ashing may be performed. The ashing may improve the adhesiveness between the organic barrier layer **14** and the second inorganic barrier layer **16**. Namely, the ashing may be used to modify (make hydrophilic) the surface of the organic barrier layer **14**, as well as to remove an extra portion of the organic barrier layer once formed.

(131) The ashing may be performed by use of a known plasma ashing device, a known photoexcitation ashing device, or a known UV ozone ashing device. For example, plasma ashing using at least one type of gas among N₂O, O₂ and O₃, or a combination of such plasma ashing and ultraviolet irradiation, may be performed. In the case where an SiN_x film is formed by CVD as each of the first inorganic barrier layer **12** and the second inorganic barrier layer **16**, N₂O is used as a material gas. Therefore, use of N₂O for the ashing provides an advantage that the device is simplified.

(132) In the case where the ashing is performed, the surface of the organic barrier layer **14** is oxidized and thus is modified to be hydrophilic. In addition, the surface of the organic barrier layer **14** is shaved almost uniformly, and extremely minute convexed and concaved portions are formed, which increases the surface area size. The effect of increasing the surface area size provided by the ashing is greater for the surface of the organic barrier layer **14** than for the first inorganic barrier layer **12** formed of an inorganic material. Since the surface of the organic barrier layer **14** is modified to be hydrophilic and the surface area size of the surface is increased, the adhesiveness of the organic barrier layer **14** with the second inorganic barrier layer **16** is improved.

(133) After the above, the resultant body is transported to a CVD chamber in order to form the second inorganic barrier layer **16**. The second inorganic barrier layer **16** is formed under, for example, the same conditions as those for the first inorganic barrier layer **12**. The second inorganic barrier layer **16** is formed in the region where the first inorganic barrier layer **12** is formed. Therefore, the inorganic barrier layer joint portion, where the first inorganic barrier layer **12** and the second inorganic barrier layer **16** are in direct contact with each other, is formed in the non-solid portion of the organic barrier layer **14**. For this reason, as described above, water vapor in the air is suppressed or prevented from reaching the inside of the active region via the organic barrier layer.

(134) The first inorganic barrier layer **12** and the second inorganic barrier layer **16** are formed, for example, as follows. An inorganic barrier layer having a thickness of 400 nm may be formed by plasma CVD using SiH₄ gas and N₂O gas, at a film formation rate of 400 nm/min, in a state where, for example, the temperature of the substrate on which the inorganic barrier layer is to be formed (the temperature of the OLED **3**) is controlled to be 80° C. or lower. The inorganic barrier layer thus formed has a refractive index of 1.84 and a transmittance of 90% to visible light having a wavelength of 400 nm (thickness: 400 nm). The inorganic barrier layer has a film stress having an absolute value of 50 MPa.

(135) The inorganic barrier layer may be an SiO₂ layer, an SiO_xN_y (x>y) layer, an SiN_xO_y (x>y) layer, an Al₂O₃ layer or the like as well as an SiN_x layer. Examples of the photocurable resin include vinyl group-containing monomers. Among the vinyl group-containing monomers, an acrylic monomer is preferably used. A photoinitiator may be incorporated into the acrylic monomer when necessary. As the acrylic monomer, any of various known acrylic monomers is usable. A plurality of acrylic monomers may be mixed together. For

example, a two-functional monomer and a monomer including three or more functional groups may be mixed together. An oligomer may be mixed. The photocurable resin, before being cured, has a viscosity at room temperature (e.g., 25°C) that is preferably lower than, or equal to, 10 Pa·s, and is especially preferably 1 to 100 mPa·s. If the viscosity is too high, it may be difficult to form a thin film having a thickness of 500 nm or less.

(136) In the above, embodiments of an OLED display device including a flexible substrate and a method for producing the same are described. Embodiments of the present invention are not limited to those described above. Embodiments of the present invention are widely applicable to an organic EL device (e.g., organic EL illumination device) including an organic EL element formed on a substrate that is not flexible (e.g., glass substrate) and a thin film encapsulation structure formed on the organic EL element.

INDUSTRIAL APPLICABILITY

(137) An embodiment of the present invention is applicable to an organic EL device and a method for producing the same. An embodiment of the present invention is especially preferably applicable to a flexible organic EL display device and a method for producing the same.

REFERENCE SIGNS LIST

(138) **1** substrate (flexible substrate) **2** backplane (circuit) **2Pa** inorganic protective layer (first inorganic protective layer), inorganic protective film (first inorganic protective film) **2Pa1** first inorganic protective layer, first inorganic protective film **2Pa2** second inorganic protective layer, second inorganic protective film **2Pb** organic flattening layer, organic flattening film **3** organic EL element **4** polarizing plate **10** thin film encapsulation structure (TFE structure) **12** first inorganic barrier layer (SiN.sub.x layer) **14** organic barrier layer (acrylic resin layer) **16** second inorganic barrier layer (SiN.sub.x layer) **20** element substrate **26** acrylic monomer **26p** vapor of acrylic monomer or mist-like acrylic monomer **100**, **100A**, **100D** organic EL display device **200** film formation device

Claims

1. An organic electroluminescent device, comprising: a substrate; a driving circuit layer including: a plurality of thin-film transistors (TFTs) formed on the substrate, a plurality of gate bus lines and a plurality of source bus lines, each of the plurality of gate bus lines and each of the plurality of source bus lines connected with one of the plurality of TFTs, a plurality of terminals, and a plurality of lead wires, each connecting one of the plurality of terminals to one of the plurality of gate bus lines or to one of the plurality of source bus lines; a first inorganic protective layer formed on the driving circuit layer and exposing at least the plurality of terminals; an organic flattening layer formed on the first inorganic protective layer; a second inorganic protective layer formed on the organic flattening layer and having a surface that has an arithmetic average roughness of 50 nm or less; an organic electroluminescent element layer, including a plurality of organic electroluminescent elements, formed on the second inorganic protective layer, and each of the plurality of organic electroluminescent elements connected with either one of the plurality of TFTs; and a thin film encapsulation structure formed, such that to cover the organic electroluminescent element layer and including a first inorganic barrier layer, an organic barrier layer in contact with a top surface of the first inorganic barrier layer, and a second inorganic barrier layer in contact with a top surface of the organic barrier layer, the organic barrier layer being formed in a region enclosed by an inorganic barrier layer joint portion where the first inorganic barrier layer and the second inorganic barrier layer are in direct contact with each other, wherein: as viewed in a direction of normal to the substrate, the organic flattening layer is formed in a region where the first inorganic protective layer is formed, the plurality of organic electroluminescent elements is located in a region where the organic flattening layer is formed, an outer perimeter of the thin film encapsulation structure crosses the plurality of lead wires and is present between an outer perimeter

- of the organic flattening layer and an outer perimeter of the first inorganic protective layer, and in a portion where the first inorganic protective layer and the first inorganic barrier layer are in direct contact with each other on the plurality of lead wires, a side surface of a cross-sectional shape of the first inorganic barrier layer, that is parallel to a line width direction of the plurality of lead wires, has a tapering angle of less than 90 degrees.
2. The organic electroluminescent device of claim 1, wherein the tapering angle of the side surface of the first inorganic barrier layer is less than 70 degrees.
 3. The organic electroluminescent device of claim 1, wherein the organic flattening layer is formed of a non-photosensitive resin.
 4. The organic electroluminescent device of claim 1, wherein the organic flattening layer is formed of polyimide.
 5. A method for producing the organic electroluminescent device of claim 1, the method comprising: step A of forming the driving circuit layer on the substrate; step B of forming a first inorganic protective film on the driving circuit layer; step C of forming an organic flattening film on the first inorganic protective film; step D of forming a second inorganic protective film on the organic flattening film; step E of treating a surface of the second inorganic protective film with chemical mechanical polishing; step F of patterning the first inorganic protective film, the organic flattening film, and the second inorganic protective film to form the first inorganic protective layer, the organic flattening layer, and the second inorganic protective layer; step G of baking the organic flattening layer to a temperature of 100° C. or higher; and step H of, after the step G, forming an organic layer on the second inorganic protective layer, the organic layer including the plurality of organic electroluminescent elements.
 6. The method of claim 5, wherein the step F includes the step of forming a contact hole running through the first inorganic protective film, the organic flattening film, and the second inorganic protective film in one step.
 7. The method of claim 5, further comprising: after the step F but before the step G, step F1 of forming a positive photoresist film covering the organic flattening layer; and step F2 of exposing an entirety of the positive photoresist film to light and then developing the positive photoresist film to remove the positive photoresist film.
 8. The method of claim 7, further comprising: the step of, between the step F1 and the step F2, storing or transporting the substrate having the positive photoresist film formed thereon.
 9. The method of claim 5, further comprising: step I of, after the step H, forming the first inorganic barrier layer selectively on an active region where the plurality of organic electroluminescent elements is formed; step J of, after the step I, locating the substrate in a chamber and supplying vapor of a photocurable resin or a mist-like photocurable resin into the chamber; step K of condensing the photocurable resin on the first inorganic barrier layer such that the photocurable resin is not present on a portion, of the first inorganic barrier layer, that has the tapering angle less than 90 degrees; and step L of, after the step K, irradiating the condensed photocurable resin with light to form the organic barrier layer of a photocured resin.
 10. The method of claim 5, further comprising: step I of, after the step H, forming the first inorganic barrier layer selectively on an active region where the plurality of organic electroluminescent elements is formed; step J of, after the step I, locating the substrate in a chamber and supplying vapor of a photocurable resin or a mist-like photocurable resin into the chamber; step K of condensing the photocurable resin on the first inorganic barrier layer to form a liquid film; step L of irradiating the liquid film of the photocurable resin with light to form a photocured resin layer; and step M of partially ashing the photocured resin layer to form the organic barrier layer.
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